

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA0615	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
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Section / Batch:	B. Tech AI&DS	Date:	16-07-2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1. Application: Weather Forecasting Simulation System

A weather simulation system uses two algorithms with complexities $O(n \log n)$ and $O(n^2 \log n)$. Analyze both algorithms using asymptotic notation and compare their growth rates. Explain how each algorithm performs as the dataset size increases. Determine which algorithm is more efficient for large-scale simulations. Justify your answer based on computational efficiency and scalability.

Application : Weather Forecasting Simulation System

Given

$$\text{Algorithm A} = O(n \log n)$$

$$\text{Algorithm B} = O(n^2 \log n)$$

Step-1 : Asymptotic Analysis

Algorithm A

$$T(n) = O(n \log n)$$

Growth rate :

$$\rightarrow \text{Linear term} = n$$

$$\rightarrow \text{Logarithmic term} = \log n$$

$$\text{Overall complexity} : O(n \log n)$$

Algorithm

$$T(n) = O(n^2 \log n)$$

Growth rate

$$\rightarrow \text{Quadratic term} = n^2$$

$$\rightarrow \text{Logarithmic term} = \log n$$

$$\text{Overall complexity} : O(n^2 \log n)$$

Step 2 : Comparison

Dataset Size(n)	$O(n \log n)$	$O(n^2 \log n)$
Small	Fast	Acceptable
Medium	Fast	Much Slow
Large	Efficient	Very Slow

Ex : For $n = 1000$

$$n \log_2 n \approx 1000 \times 10 = 10,000$$

$$n^2 \log_2 n \approx 1000^2 \times 10 = 10,000,000$$

Algorithm B performs about 1000 times more work.

Step 3: Growth Rate

As n increases

→ $O(n \log n)$ grows slowly

→ $O(n^2 \log n)$ grows rapidly because of the quadratic term.

Therefore,

$$O(n \log n) < O(n^2 \log n)$$

Step 4: Efficiency and Scalability

→ Algorithm A requires much less computation

→ Uses less processing time

→ Suitable for large weather datasets

→ Scales efficiently as data size increases.

Q2. Application: Online Learning Recommendation System

An online learning platform processes user interaction data using the recurrence $T(n) = T(n-1) + n$. Apply the substitution method to expand the recurrence step-by-step and derive the final time complexity. Clearly show intermediate steps and simplify the expression. Analyze how the algorithm performs when the number of users grows rapidly. Interpret the efficiency using asymptotic notation and discuss whether the system is scalable for large educational platforms.

Application : Online learning Recommendation System

Given Recurrence

$$T(n) = T(n-1) + n$$

Base case :

$$T(1) = 1$$

Step 1 : Expand using Substitution Method

$$T(n) = T(n-1) + n$$

substitute $T(n-1)$

$$= T(n-2) + (n-1) + n$$

substitute again

$$= T(n-3) + (n-2) + (n-1) + n$$

continue

$$= T(1) + 2 + 3 + 4 + \dots + n$$

Step 2 : Sum of first n Natural Numbers

$$2 + 3 + \dots + n = \frac{n(n+1)}{2} - 1$$

Therefore,

$$T(n) = 1 + \left(\frac{n(n+1)}{2} - 1 \right)$$

$$T(n) = \frac{n(n+1)}{2}$$

$$T(n) = \frac{n^2 + n}{2}$$

Step 3: Time Complexity

Ignoring constants and lower order terms

$$T(n) = O(n^2)$$

Step 4: Interpretation

As the NO. Of users increases,

- processing time increases quadratically
- Doubling the users results in approximately four times the computation
- large educational platforms will experience slower recommendation generation

Step 5: scalability

Since the algorithm has

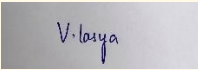
$$O(n^2)$$

complexity,

- It is suitable only for small or medium datasets
- For millions of users, performance degrades significantly
- More efficient algorithm such as $O(n \log n)$ or $O(n)$ are preferred for large-scale recommendation systems.

Code of Conduct

I V.Lasya(192524186) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action

Signature of the Student: 

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____